A Robust Co-Training Framework to Handle Noisy Labels for Remote Sensing Image Segmentation

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***Abstract*— Effective learning with noisy labels is crucial in remote sensing (RS) data analysis, not only for instancewise prediction but also for pixelwise prediction. The recent surge in demand for robust machine learning models emphasizes the necessity of constructing large datasets and managing associated costs carefully. This trend highlights the potential of using coarsely annotated datasets, which commonly entail the risk of producing noisy labels. To address label noise, methods of learning with noisy labels have been proposed for instancewise classification assessed in computer vision. However, previous studies have not provided a detailed discussion on how to detect human error at the pixel level and how to adapt the resulting error maps within a single end-to-end learning framework in remote sensing. To address these issues, we propose a new co-training framework for segmentation modeling with automatic filtering to reduce the negative effects caused by noisy labels in the training dataset. Our method has three key contributions: 1) a trainable filter module to identify uncertain pixels at each iteration; 2) a weighted multiloss strategy to effectively filter out uncertain training loss; and 3) a collaborative filter update based on outputs from co-trained networks. To assess the effectiveness of our method, we conducted experiments on the DeepGlobe land-cover classification dataset, a self-constructed mangrove dataset, and the Massachusetts building dataset. We compared the performance of our method to several state-of-the-art noise-robust methods. The experimental results reveal that our proposed method outperforms the alternatives and demonstrates greater effectiveness on multiclass segmentation tasks.**

***Index Terms*— Co-training, deep learning (DL), multiclass classification, noisy labels, remote sensing (RS), semantic segmentation.**

I. Introduction

WHILE deep learning (DL) in remote sensing (RS) data analysis has emerged as a cornerstone technique with significant expectations for practical usage such as agriculture [1], forestry [2], and urban planning [3]. The rapid increase in demand of DL has lifted the model's capacity in high-accuracy prediction and promoted careful cost management for annotation to create large datasets. This trend has led to the proliferation of experimental datasets, but it has also introduced the potential risk of noisy labels due to the tradeoff between the quality and quantity of datasets, which can commonly degrade the model's performance significantly. The causes of label noise can be roughly categorized into two factors: coarse annotation through automatic processing or crowd-sourcing, and difficulty in annotation even for experts due to the high intraclass diversity and interclass similarity in remote sensing imagery. The first factor is due to the high demand for low-cost data creation methods that introduce noises into labels, which is similar to that in general classification tasks [4]. In handling this challenging factor, one-by-one data cleansing for corrupted labels is impractical [5], and standard regularization techniques cannot completely solve this issue [4], [6]. Therefore, it is a critical task to develop robust learning with coarsely annotated datasets [7], [8], [9], [10], [11], [12]. On the other hand, the difficulty in annotation is more specific to remote sensing data analysis, making human annotation a challenging task. Many studies have tried to mitigate the negative effects of this challenge in various kinds of projects, e.g., satellite and unmanned aerial vehicle (UAV) image vegetation analysis [13], [14], [15], [16], [17], road and building detection [18], [19], multilabel image classification [20], and hyperspectral image classification [5]. These two nonnegligible factors are sensational to the development of high-quality DL models without the high-cost requirement of plenty of expert annotations.

Handling noisy labels has become one of the key tasks in robust DL. Due to the unsatisfactory results of standard regularization methods such as data augmentation, weight decay, dropout, and batch normalization [4], there has emerged a need to develop new learning paradigms optimized for robust learning with noisy labels. The initial approach to address this need was based on fine-tuning techniques, but it was found to exhibit lower accuracy compared with models trained from scratch with clean and massively noisy labels [21]. As a consequence, the research spotlight shifted to efficient training with single datasets, not considering being given clean and noisy datasets separately. As the next step, some research focused on the estimation of the latent transition matrix of labels. Patrini et al. [22] introduced an easy way to approximate it using simple datasets with synthetic noisy labels. However, it is also pointed out that real noise contains more randomness and complex patterns than synthetic ones, making it difficult to incorporate the latent transition matrix into more complex situations [23], [24], [25], especially when dealing with a large number of classes [10]. Moreover, many methods were proposed based on various types of techniques such as robust

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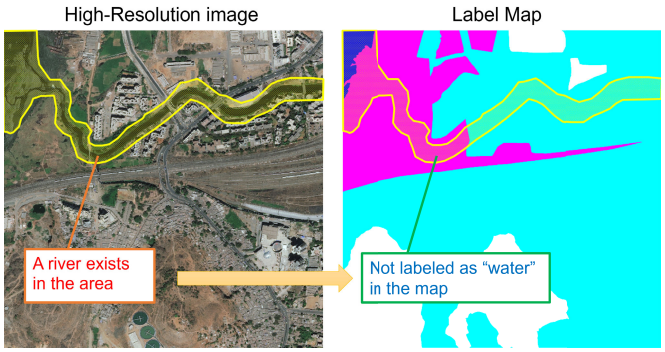


Fig. 1. Sample with label noise in the Deepglobe dataset, where the defined classes of "urban land," "rangeland," "water," and "barren land" are represented by cyan, magenta, blue, and white, respectively. The yellow enclosed area shows a misannotation where the river, which should be classified as "water," is mistakenly labeled as "urban land" or "rangeland."

network architecture [26], prominent loss functions [27], meta-learning [28], and metric learning [29], [30]. But the most popular and promising one seems to be data selection [9], [10], [11], [12], which eliminates uncertain samples with a relatively high loss function, considered as noisy ones, from datasets during training. For instance, MentorNet [9] and co-teaching [10] adopt a collaborative learning approach with two networks during training, showing a significant increase in accuracy in simulated datasets. Sample selection provides a fundamental scheme in robust learning and is referenced in other more specific fields of study [19], [24].

Despite various studies on learning with noisy labels in image classification, few studies are addressing noisy pixels in label masks for semantic segmentation. Whereas previous image classification methods often assume that labels can be classified as either clean or noisy, most label masks in semantic segmentation contain a mix of clean and noisy pixel information, making it difficult to categorize them without arbitrary decision-making. For instance, in Fig. 1, we show a sample from a segmentation dataset where the presence of a river is not reflected in the label map. In this example, since the noisy pixels appear limited in number, the instance-level distribution of loss may be too similar to that of cleaner samples, making effective sample selection challenging. Moreover, in general, the configuration of polygons in segmentation datasets inevitably includes mislabeled boundaries between different categories, as annotation is typically not done pixel by pixel, making the definition of a "clean sample" ambiguous. Due to ambiguity, training models with noisy labels in semantic segmentation are closely related to the research field of weakly supervised semantic segmentation (WSSS), where weaker forms of supervision like image-level labels or bounding boxes are used for model training. Some studies in WSSS discuss methods for reducing noisy information from various perspectives, such as early learning [31], co-training [32], and joint loss functions [33]. However, these studies primarily focus on effectively training models with weaker supervision, and it is challenging to directly generalize these methods to other types of noisy labels. Although other studies explore various ideas, such as combining label correction and sample reweighting [34], using data with different resolutions simultaneously [35], and adapting WSSS methods to more specialized fields [36], fundamental schemes for noise-robust learning in semantic segmentation with strong generalization abilities to adapt to datasets across various research fields have not been established.

In this study, we propose a novel co-training learning framework to mitigate the negative effects of noisy pixels in multiclass semantic segmentation for remote sensing data analysis, focusing on detecting noise and leveraging the detected information to enhance network training. A co-training framework is widely used for sample selection, WSSS, and other tasks aimed at reducing error accumulation in a single network [10], [32], [37], [38]. Due to the scarcity of studies on this topic, our framework leverages effective WSSS methods and integrates them into a unified model tailored for remote sensing applications. In our method, we incorporate filtering masks that capture uncertainty information in pixelwise label maps, enabling the model to account for spatial characteristics and label ambiguity inherent in semantic segmentation datasets. Our approach comprises three key technical components.

1) *Trainable Filtering Masks:* We introduce a dynamic filtering module that generates per-batch updated filtering masks. This approach helps avoid excessive learning in the early stages when the networks have not yet acquired sufficient capacity to produce reliable outputs.
2) *Weighted Multiloss Strategy:* We propose a novel weighted multiloss format in which the filtering masks are computed to serve as effective weights for the loss functions. In addition, we integrate conditional random field (CRF)-Loss into our weighted multiloss scheme to mitigate information loss, thereby preserving the spatial context of the source images in the predicted maps [33], drawing inspiration from CRF-based post-processing techniques [39].
3) *Collaborative Refinement of Filtering Module:* We introduce a novel co-training framework in which co-trained networks not only share noisy information but also provide inputs to the filtering module, thereby facilitating the learning of consensus error regions throughout the training process.

To validate the effectiveness of our method, we conducted experiments with three types of datasets: the DeepGlobe land cover classification dataset, a self-constructed Mangrove detection dataset, and the Massachusetts building footprint segmentation dataset. We conducted model training with previous methods, including SFL [19] and AIO2 [40] which shows the effectiveness for noisy binary classification but has yet to adapt to multiclass ones.

This article is organized as follows. Section II summarizes related studies on learning methodologies with noisy labels. Technical details of the proposed method are described in Section III. Next, we discuss the experimental results from two perspectives: network dependency and comparison to other methods in Section IV. This is followed by an elaboration of the results in Section V and ablation studies in Section VI, and we conclude this work with future lines in Section VII.

II. Related Work

A. Learning With Noisy Label for Image Classification

Due to the unsatisfactory reliability of automatically collected labels, many studies in image classification have

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focused on how to conduct robust learning with noisy annotated data. The recent trend of large capacity models for obtaining more generalization ability more easily caused overfitting to noisy labels due to the memorization effect [9], which forces developers to seriously and laboriously tune the learning setting. To solve these problems, the estimation of the noise transition matrix, which shows all the probabilities of label flipping from one true class to another, was proposed and performed better than the benchmark in simulated datasets [22]. Nevertheless, real-world noise is caused in a more complex scenario, especially when the class number is large [10], [23], [25]. While many robust methods to handle noisy labels have been proposed from various approaches, the least task-specific approach seems to be sample selection because of the simple strategy that selects and eliminates noisy labels from datasets by referring to uncertainty calculated from model output and labels during training [4]. MentorNet [9] uses an additional network providing meaningful supervision not to cause serious memorization of the training model to noisy labels by referring to the curriculum learning (CL) scheme. This method adopts the small-loss trick that selects samples with low losses under the threshold in each batch operation, regarding them as clean samples, and uses only selected samples for backpropagation (self-paced MentorNet). Despite achieving success in some experiments, it has been pointed out that this method may lead to error accumulation, as the selective information of uncertain samples in one network will be directly transferred back to itself in the next batch calculation [10]. Co-teaching [10] addresses this drawback using two networks with the same structure along the co-training scheme, which is a common technique for the compensation of crucial information to each other and reducing the risk of favoring the samples in a single network [10], [37], [38], [41]. In co-teaching, each of the two networks has its parameters and can extract different types of features from each other, which enables them to filter out various errors introduced by noisy labels. From a different perspective, the multiround learning approach [11], [12] iteratively selects a set of clean samples by repeating training rounds without the necessity of training two networks with the same structure. Overall, many methods following the sample selection scheme have demonstrated effectiveness for handling noisy labels, but it is worth noting that they may suffer from incorrect selection when the loss distributions of true-labeled and false-labeled samples overlap significantly [4].

B. Weakly Supervised Semantic Segmentation

Learning with noisy labels can be regarded as one of the weakly supervised learning problems, so it is meaningful to search for adaptive technical approaches within the field of WSSS. In WSSS, pseudolabel masks generated from class activation maps are usually used for training segmentation models, instead of manually annotated ones, after the production process of well-qualified class activation maps using instancewise annotated datasets. In most cases, research in WSSS focuses on how to produce high-quality masks in the first label creation stage, but some studies also discuss the quality of training in the second learning stage with noisy labels, which originates from the unreliability of weakly annotated datasets. Li et al. [42] notes the scarcity of specific learning methods in the second stage and proposes a filtering operation to loss maps conducted by the multiplication of weight masks referring to uncertainty on the pixels. From a different viewpoint, ADELE [31] introduces "early learning" in noisy label learning, which monitors the process of evaluation index for each category to find the beginning point of overfitting and stop learning on the class, and starts pixelwise label correction if the index has already saturated. This idea is also used in building segmentation [40]. As a similar approach to image classification, Rong et al. [32] adopts the co-training framework in the WSSS, like co-teaching [10], and incorporates a boundary-enhanced strategy to boost predictions on difficult boundary areas. From the perspective of compensating for uncertain pixels, some studies pay attention to the unique idea of CRF, commonly used for postprocessing in semantic segmentation [39], which can refine the shape of predicted objects by considering relationships such as similarity and location between pixels in raw images, and introduce advanced loss functions [33], [36], [43]. Overall, some studies in WSSS have demonstrated significant insights, but their scope does not fully account for real-world noise caused by human annotators. Direct adaptation to remote sensing modeling is challenging due to the diverse causes of label noise resulting from human annotation, as highlighted by Liu et al. [40].

C. Learning With Noisy Label in Remote Sensing Data Analysis

Regarding remote sensing data analysis, label noise significantly affects model performance. Several studies have addressed this issue in image classification projects, driven by the high demand for coarsely annotated data, which is often produced by crowdsource procedures or publicly available thematic products [44], [45]. Instinctively, it may seem reasonable to follow general approaches in computer vision to tackle this issue, but it has been noted that specific causes of noise in multilabel scene classification on remote sensing images necessitate the development of methods optimized for the remote sensing field [20], [46], which motivated the development of many methods for various tasks [6], [29], [47].

In semantic segmentation, enhancing robustness to noisy labels becomes much harder than in image classification because of its spatial aspect. The common causes of noisy labels can be categorized into two cases: coarse annotation via crowdsourcing or automated annotation processes and hard annotation requiring a high level of expertise. While the former aims to reduce annotation costs, the latter originates from the intrinsic difficulty of remote sensing data analysis. Even in the same area, multiple remote sensing images captured at different times can exhibit spectral differences due to seasonal changes and weather conditions. Despite its significant influence on model performance, only a few studies have focused on the topic of noisy learning in semantic segmentation [19], [35], [40]. Dong et al. [35] conducted an experiment on how to effectively train models with noisy datasets by combining the low-resolution landcover product (10 m) and high-resolution satellite images (3 m) captured in China, proposing an online noise correction framework consisting of an uncertainty estimation module and an adaptive correction module. Song et al. [19] and Liu et al. [40] focused on the effective use of publicly available building detection datasets with incomplete and inaccurate noisy labels, referring to previous

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methods of instancewise sample selection [9], [10] and early learning [31], respectively.

III. Methodology

Following previous approaches in image classification, we leverage the co-training framework [10], [32] to achieve robust learning despite noisy labels. Because our method is designed for multiclass segmentation tasks, which are significantly more complex than image classification, we introduce a novel co-training framework tailored to these challenges, taking into account the spatial aspect of labels within a single map. To address this challenging task, we introduce three modules: the mask production module (MPM), the noise label production module (NLPM), and the loss-filtering module (LFM), each specifically tailored to our purpose. In this article, a scalar is denoted by a lowercase letter, e.g., $y \in \mathbb{R}$. A matrix, denoted by a bold capital letter, e.g., $\mathbf{Y} \in \mathbb{R}^{I \times J}$, has components Y_{ij}, where $i = 1, 2, \ldots, I$ and $j = 1, 2, \ldots, J$. A tensor is represented by a bold calligraphic letter, e.g., $\mathcal{Y} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$. Here, we use matrix form when representing 2-D maps, even where each pixel is assigned a specific index in a 1-D format, aligning with intuitive understanding. In the following, Section III-A provides a brief introduction to pixelwise co-training, and Section III-B shows the base framework of our method. The MPM and NLPM are discussed in Section III-C, and the process of filtering the loss function in the LFM is introduced in Section III-D.

A. Preliminaries

Due to the robustness to noisy information, we use the co-training framework to train models on datasets containing uncertain samples, which include both noisy and excessively difficult ones, thereby suppressing the risk of error accumulation in a single network. The initial idea is to combine this framework with a "small-loss trick" [9] where only the samples with low loss values are selected for updating models, which is widely used in image classification to handle noisy labels [9], [10], [12], [48]. In semantic segmentation, this selection process should be reformulated as a filtering procedure applied to the loss maps on the opposite side [49] due to the 2-D format of labels, where the loss value of uncertain pixels is forced to be zero. This leads to the definition of the masked cross entropy (MCE) loss function L_{MCE} as follows:

$$L^{A}_{\mathrm{MCE}} = \frac{1}{N_p} \sum_{i=1}^{N_b} \sum_{j=1}^{HW} \left(M^{B}_{ij} \cdot L_{\mathrm{CE}}\left(f(\boldsymbol{\theta}_A, \mathbf{X}_i), Y_{ij} \right) \right) \tag{1}$$

$$L^{B}_{\mathrm{MCE}} = \frac{1}{N_p} \sum_{i=1}^{N_b} \sum_{j=1}^{HW} \left(M^{A}_{ij} \cdot L_{\mathrm{CE}}\left(f(\boldsymbol{\theta}_B, \mathbf{X}_i), Y_{ij} \right) \right) \tag{2}$$

where L_{CE} represents the cross-entropy (CE) loss function, which includes the last activation layer such as Sigmoid or Softmax functions. The filtering mask $M^{A(B)}_{ij}$ is defined as

$$M^{A(B)}_{ij} = \begin{cases} 1, & \text{if } L_{\mathrm{CE}}\left(f(\theta_{A(B)}, \mathbf{X}_i), Y_{ij} \right) \leq \tau \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

where f represents the segmentation network, N_p is the number of all the unmasked pixels, N_b is the number of batch instances, and $\mathbf{X}_i$ and $\mathbf{Y}_i$ are an images and a label mask of the ith sample in the mini-batch. $\boldsymbol{\theta}_A$ and $\boldsymbol{\theta}_B$ represent the parameters of Networks A and B, respectively, and τ is the threshold for pixelwise sample selection.

Besides this formulation, there are significant learning risks during training, which differ from those in image classification. First, it is challenging to categorize semantic segmentation label masks into clean or unclean groups since a label map inevitably contains ambiguous labels, especially in boundary regions. Second, the training process contains the risk of causing unstable learning because each pixel also contributes to constructing a single map with complex spatial information; excessive filtering may result in fragmented masks lacking sufficient detailed spatial information.

B. Base Framework

To reduce these risks, we design novel modules to optimize how to detect noisy pixels and weaken their contribution, thereby mitigating their negative impact on model training. In our framework, we introduce three new modules—MPM, LFM, and NLPM—, as illustrated in Fig. 2.

The MPM is designed to detect label noise and generate filtering masks consisting of continuous values, allowing for more flexible and fine-grained filtering, unlike the discrete ones described in (3). If we represent a set of logits as $\mathcal{Z}^{A(B)} = \{\mathbf{Z}^{A(B)}_1, \mathbf{Z}^{A(B)}_2, \ldots, \mathbf{Z}^{A(B)}_{N_b}\}$ where each logit is computed as $\mathbf{Z}^{A(B)}_i = f(\theta_{A(B)}, \mathbf{X}_i)$, a set of label masks as $\mathcal{Y} = \{\mathbf{Y}_1, \mathbf{Y}_2, \ldots, \mathbf{Y}_{N_b}\}$, and a set of filter masks $\mathcal{M} = \{\mathbf{M}_1, \mathbf{M}_2, \ldots, \mathbf{M}_{N_b}\}$, we can define the MPM as follows:

$$\mathcal{M}^{A(B)} = \mathrm{MPM}\left(\mathcal{Z}^{A(B)}, \mathcal{Y}, \boldsymbol{\theta}^{\mathrm{MPM}} \right). \tag{4}$$

The MPM is a trainable module with weights $\boldsymbol{\theta}^{\mathrm{MPM}}$ to learn the optimized mask for filtering the loss function on the opposite side, differing from those in (3). To update the MPM, we introduce the NLPM module, which uses a hyperparameter α to effectively generate the "noise" label masks $\mathcal{N} = \{\mathbf{N}_1, \mathbf{N}_2, \ldots, \mathbf{N}_{N_b}\}$ where $\mathbf{N}_i$ represents the noise mask of the ith sample

$$\mathcal{N}^{A(B)} = \mathrm{NLPM}\left(\mathcal{Z}^{A(B)}, \mathcal{Y}, \alpha \right). \tag{5}$$

We then update the parameters using simple CE-Loss calculations

$$L^{A \to B}_{\mathrm{NEL}} = \frac{1}{N_p} \sum_{i=1}^{N_b} \sum_{j=1}^{HW} L_{\mathrm{CE}}\left(M^{A}_{ij}\left(\boldsymbol{\theta}^{\mathrm{MPM}}; \mathcal{Z}^{A}, \mathcal{Y} \right), N^{B}_{ij} \right) \tag{6}$$

$$L^{B \to A}_{\mathrm{NEL}} = \frac{1}{N_p} \sum_{i=1}^{N_b} \sum_{j=1}^{HW} L_{\mathrm{CE}}\left(M^{B}_{ij}\left(\boldsymbol{\theta}^{\mathrm{MPM}}; \mathcal{Z}^{B}, \mathcal{Y} \right), N^{A}_{ij} \right) \tag{7}$$

where $L^{A \to B(B \to A)}_{\mathrm{NEL}}$ is defined as the noise estimation loss (NEL) for updating the parameters $\boldsymbol{\theta}^{\mathrm{MPM}}$ in filtering mask $M^{A(B)}_{ij}$, which estimates noisy pixels in the side B(A) referring to the prediction results and label map in side A(B).

As shown in (6) and (7), the training of the MPM follows a collaborative framework, where the output from the opposite side is referenced during training, aligning with a co-training structure. This format mitigates biases from a single side, leading to optimized mask generation throughout the learning process, as discussed in detail in Section III-C. Consequently,

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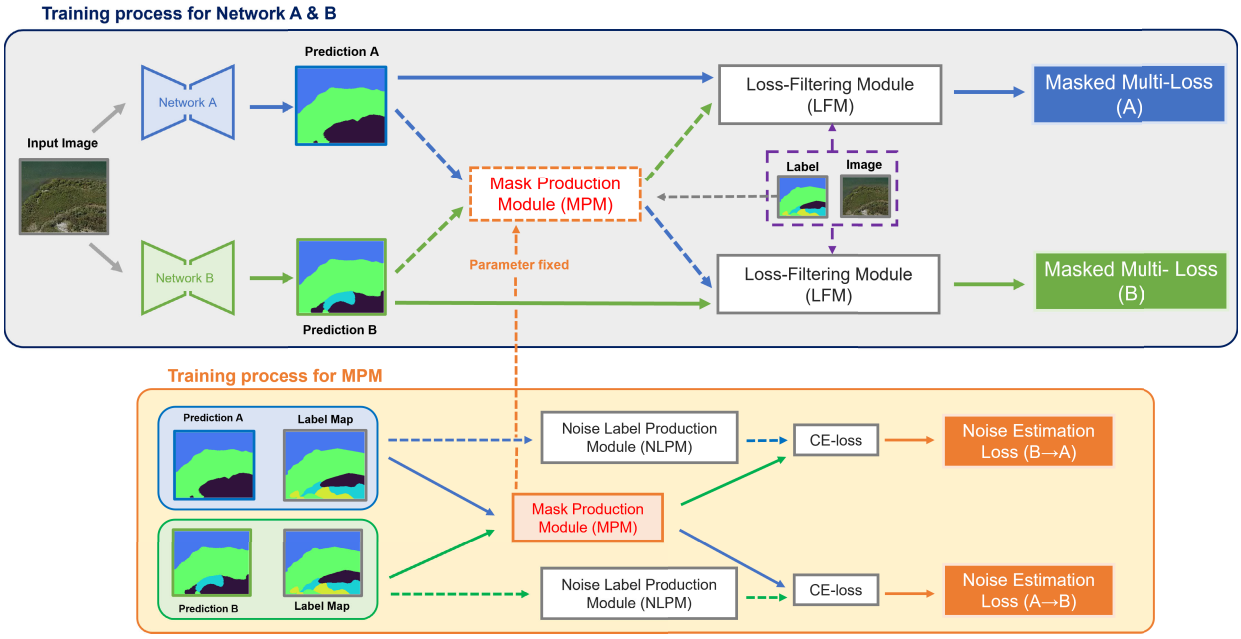


Fig. 2. Overview of the proposed framework. Blue and green lines represent the propagation flows from Network A and Network B, respectively. Dotted lines indicate forward propagation paths, while solid lines represent flows used for parameter updates during backpropagation. The notation A → B (or B → A) indicates the prediction direction, where the MPM receives predictions from Network A (or B) and predicts noise pixels on the opposite side B (or A).

the two losses can be integrated into a single formulation for clarity, as follows:

$$L_{\mathrm{NEL}} = \frac{1}{2}\left(L_{\mathrm{NEL}}^{A\to B} + L_{\mathrm{NEL}}^{B\to A}\right). \tag{8}$$

Next, we discuss how to incorporate filtering masks $\mathcal{M}$ with noise information into the training of the two networks. In the mask production process, the filtering mask $\mathbf{M}$ is defined to include continuous values in the range [0, 1], allowing it to be used for loss filtering, as discussed previously in (1) and (2). However, it remains unclear whether the losses associated with uncertain pixels should be completely disregarded during training, particularly when these pixels may contain valuable information related to surrounding or visually similar objects. Moreover, consideration should be given to including other loss functions besides CE-Loss, as multiclass classification commonly incorporates Dice or Jaccard coefficient loss to deal with class imbalance.

To meet these requirements, we define an additional module, LFM, to effectively incorporate a filtering operation into the multiloss format, so (1) and (2) can be changed to the following masked multiloss (MML) with hyperparameter β and w:

$$L_{\mathrm{MML}}^{A(B)} = \mathrm{LFM}(\mathcal{M}^{B(A)}, f(\boldsymbol{\theta}_{A(B)}, \mathcal{X}), \mathcal{X}, \mathcal{Y}, \beta, w). \tag{9}$$

Equation (9) includes a set of input images $\mathcal{X}$ and label masks $\mathcal{Y}$, because we incorporate CRF-Loss [33] into our framework, as explained in detail in Section III-D.

In the end, we can complete defining the optimization process through batch calculations using (6)–(9). The calculation proceeds in two steps: the first step updates the parameters of MPM, and the second updates the two networks. These two optimization processes can be described as follows:

$$\hat{\boldsymbol{\theta}}^{\mathrm{MPM}} = \underset{\boldsymbol{\theta}^{\mathrm{MPM}}}{\operatorname{argmin}} L_{\mathrm{NEL}}\left(\boldsymbol{\theta}^{\mathrm{MPM}}; \mathcal{Z}^A, \mathcal{Z}^B, \mathcal{N}^A, \mathcal{N}^B, \mathcal{Y}, \alpha\right) \tag{10}$$

$$\hat{\boldsymbol{\theta}}^{A} = \underset{\boldsymbol{\theta}^{A}}{\operatorname{argmin}} L_{\mathrm{MML}}^{A}\left(\boldsymbol{\theta}^{A}; \mathcal{M}^B, \mathcal{X}, \mathcal{Y}, \beta, w\right) \tag{11}$$

$$\hat{\boldsymbol{\theta}}^{B} = \underset{\boldsymbol{\theta}^{B}}{\operatorname{argmin}} L_{\mathrm{MML}}^{B}\left(\boldsymbol{\theta}^{B}; \mathcal{M}^A, \mathcal{X}, \mathcal{Y}, \beta, w\right) \tag{12}$$

where $\hat{\boldsymbol{\theta}}^{\mathrm{MPM}}$, $\hat{\boldsymbol{\theta}}^{A}$, and $\hat{\boldsymbol{\theta}}^{B}$ represent the optimized parameters of MPM, Network A, and Network B, respectively.

C. Mask Production Process

The two modules, MPM and NLPM, are designed to perform noise detection and generate appropriate masks to enhance network updates. As demonstrated in (4)–(7), MPM learns parameters to generate suitable mask values for filtering the loss map on the opposite side, with training facilitated by sample masks computed from NLPM. This unique structure can be understood as an additional collaborative learning mechanism as shown in Fig. 2, where predictions from both the co-trained networks work together to effectively optimize the parameters of MPM, thereby preventing the accumulation of model uncertainty from a single network.

In addition, the collaborative framework is expected to automate the warm-up process [10]. The warm-up process is a technique that limits loss reduction until the networks have been trained sufficiently to produce reliable outputs, thereby ensuring reasonable information selection. In this collaborative process, the MPM is trained to detect only the predictable errors in each network in an alternative manner while preserving individual learning. During the initial learning stage, the NLPM may not produce meaningful sample masks due to limited accuracy in the co-trained network, resulting in

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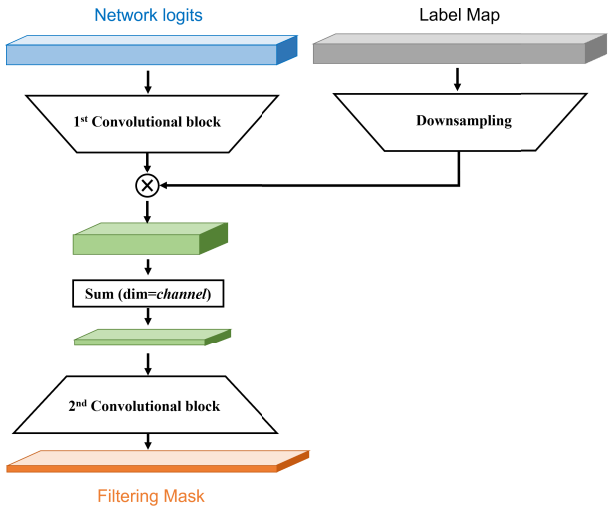


Fig. 3. Overview of the MPM.

low-confidence filtering masks in the MPM. In contrast, once the networks have achieved reasonable accuracy, the MPM extracts patterns from pairs of network outputs and label maps, generating filtering masks with higher confidence values and effectively filtering out noisy or overly challenging pixels. This learning process is expected to enable a smoother transition to the noise filtering stage.

In Fig. 3, we illustrate the overview of the network structure of MPM. The input to this module is network logits and label maps. The module can be divided into two convolutional blocks and a merging process. The first convolutional block is designed to extract relationships between local features using a single convolutional layer with a kernel size of 5, a stride corresponding to the scale factor, and a hyperbolic tangent activation function [50]. Following this, a merging process, consisting of multiplication and summation operation, captures the relationships between local features and labels, considering label flipping similar to the approach of the transition matrix [22]. Here, the hyperbolic tangent function produces values in the range $[-1, 1]$, ensuring that high-confidence values on wrongly classified pixels are not ignored during the multiplication operation with one-hot label vectors unlike the Softmax function, which would yield values close to zero. The second convolutional block aims to capture spatial relationships on a larger scale. It includes two convolutional layers with a kernel size of 3 to spatially smooth feature maps, followed by a Sigmoid layer and an upsampling layer to recover the original size and produce the final masks.

With regard to NLPM, we incorporate a simple statistical calculation to generate a noise label map used as a sample mask for updating MPM. We assume that noisy labels contribute to significantly higher loss values compared with clean ones during batch calculations and establish a threshold based on statistical measures of the mean and standard deviation, which allows us to identify pixels with high loss values. Our setting can be represented as follows:

$$N_{ij}^{A(B)} = \begin{cases} 1, & \text{if } L_{\text{CE}}\left(Z_{ij}^{A(B)}, Y_{ij}\right) \leq \mu + \alpha \times \sigma \\ 0, & \text{otherwise} \end{cases} \tag{13}$$

where α is a hyperparameter depending on the feature of each dataset, and μ and σ are defined as

$$\mu = \frac{1}{N_b} \sum_{i=1}^{N_b} \sum_{j=1}^{HW} L_{\text{CE}}\left(Z_{ij}^{A(B)}, Y_{ij}\right)$$

$$\sigma = \sqrt{\frac{1}{N_b} \sum_{i=1}^{N_b} \sum_{j=1}^{HW} \left(L_{\text{CE}}\left(Z_{ij}^{A(B)}, Y_{ij}\right) - \mu\right)^2}.$$

The statistical calculation is designed to enable NLPM to dynamically produce either 1 or 0, depending on the mini-batch sample selection. In our framework, we consider two scenarios that can lead to particularly high-loss pixels: 1) boundary gaps caused by the coarse configuration of target polygons and 2) label flipping. The first scenario, boundary gaps, should be detected through statistical operations when the predicted maps are sufficiently accurate at the boundaries. On the other hand, the second scenario, label flipping, presents challenges due to annotation difficulties in remote sensing. NLPM reflects this uncertainty by leveraging statistical calculations over the batch. When the loss in the labeled regions is relatively high, the noise label is more likely to be assigned a value of 0; otherwise, it is more likely to be assigned a value of 1. This selection-dependent approach is expected to reflect the ambiguity of noise labels in the confidence values of filtering masks through MPM training, stabilized by the mutual parameter updates within the collaborative learning framework.

D. Filtering Process to Loss Function

To effectively filter out loss values on noisy pixels, we introduce an LFM that guides the incorporation of filtering masks into multiloss functions. The "small-loss" trick [9] typically uses a single CE-Loss for data selection. However, incorporating multiloss with either Dice loss or Jaccard loss is a common strategy for multiclass segmentation tasks. In addition, we account for the existence of filtered-out pixels by introducing CRF-Loss to the multiloss to reuse the spatial information of the input image, referencing previous WSSS studies [33]. The CRF-Loss compensates for insufficient information in weakly annotated masks by facilitating interactions between nearby pixels. Given the popularity of postprocessing based on CRF, it intuitively seems to also be effective for noisy information. However, transferring information from pixel to pixel on predicted maps carries unignorable risks, particularly when it involves unreliable pixels over large areas; this might propagate noisy information to clean pixels. To solve these issues, we need to establish an asymmetrical flow between "clean" and "noisy" pixels, allowing a flow from clean to noisy, while filtering out the opposite direction.

Regarding the filtering process in CE-Loss, we can directly use it as weights, as discussed in Section III-A. However, this approach clearly involves the risk of overfiltering, which may result in the loss of useful information. To address this issue, we have made slight adjustments by introducing a hyperparameter β defined in $[0, 1]$ to prevent excessive filtering and avoid significant reduction of information in

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datasets, as follows:

$$L^{A}_{\mathrm{MCE}} = \frac{1}{N_p}\sum_{i=1}^{N_b}\sum_{j=1}^{HW}\left(\hat{M}^{B}_{ij}\cdot L_{\mathrm{CE}}\left(f(\boldsymbol{\theta}_A, \mathbf{X}_i), Y_{ij}\right)\right) \tag{14}$$

$$L^{B}_{\mathrm{MCE}} = \frac{1}{N_p}\sum_{i=1}^{N_b}\sum_{j=1}^{HW}\left(\hat{M}^{A}_{ij}\cdot L_{\mathrm{CE}}\left(f(\boldsymbol{\theta}_B, \mathbf{X}_i), Y_{ij}\right)\right) \tag{15}$$

where

$$\hat{M}^{A(B)}_{ij} = (1-\beta) + \beta\cdot M^{A(B)}_{ij}.$$

As mentioned earlier, we incorporate CRF-Loss [33] with some modifications to compensate for information on unlabeled pixels. To prevent the negative propagation of predicted results, we also use filtering masks from MPM, enabling an asymmetric flow of information from clean to noisy pixels. For CRF-loss calculation, we define the energy formulation between pixels i and j, following [33]

$$E(i,j) = G(i,j)\sum_{c\in C}\left(P^{c}_{\mathrm{net}}(i)\left(1-P^{c}_{\mathrm{net}}(j)\right)\right). \tag{16}$$

Here, P^{c}_{net} represents the softmax output of the segmentation network at pixels i and j for class c. $G(i,j)$ is a Gaussian kernel bandwidth filter, defined as

$$G(i,j) = w\exp\left(-\frac{||D_i - D_j||^2}{2\sigma_d^2} - \frac{||I_i - I_j||^2}{2\sigma_r^2}\right) \tag{17}$$

where w is the weight parameter, D represents the pixel spatial position, and I is the RGB color. σ_d and σ_r are the scale parameters of Gaussian kernels, controlling the smoothness and influence of pixel intensity, respectively.

To prevent the undesired update of outputs on clean pixels within the context of uncertain pixels, we introduce a detached output variable $\bar{P}_{\mathrm{net}}$, derived from P_{net}. The energy term $E(i,j)$ in (16) is consequently redefined to incorporate the detached output on site j as $E(i,\bar{j})$. In addition, we define the weight masks W^{NN}_{ist} and W^{CN}_{ist} as follows:

$$W^{\mathrm{NN}}_{ist} = (1-M_{is})(1-M_{it})$$
$$W^{\mathrm{CN}}_{ist} = (1-M_{is})M_{it}.$$

Here, W^{NN}_{ist} and W^{CN}_{ist} are associated with the probability of selected pixels being a noise–noise pair and a clean–noise pair, respectively. From these definitions, our weighted CRF-Loss (WCRF-loss) is reformulated from standard one [33] as follows:

$$L_{\mathrm{WCRF}} = \frac{2}{N_b}\sum_{i=1}^{N_b}\sum_{(s,t)\in\Omega}\left(W^{\mathrm{NN}}_{ist}\cdot E_i(s,t) + W^{\mathrm{CN}}_{ist}\cdot E_i(s,\bar{t})\right) \tag{18}$$

where Ω represents the paired set of pixels selected by the CRF kernel.

To address the data imbalance, we directly incorporate the Dice loss L_{Dice} into our framework without any filtering operations. This is because this type of loss function conducts nonpixelwise calculations by referencing the total output, making it less susceptible to changes caused by a small percentage of noisy pixels in batch samples.

From these definitions of masked loss functions, we can obtain the format of LFM as follows:

$$\begin{aligned}\mathrm{LFM}(\mathcal{M}, f(\boldsymbol{\theta},\mathcal{X}),\mathcal{X},\mathcal{Y},\beta,w) &= L_{\mathrm{MCE}}(\mathcal{M}, f(\theta,\mathcal{X}),\mathcal{Y},\beta)\\ &\quad + L_{\mathrm{WCRF}}(\mathcal{M}, f(\theta,\mathcal{X}),\mathcal{X},\mathcal{Y},w)\\ &\quad + L_{\mathrm{Dice}}(f(\theta,\mathcal{X}),\mathcal{Y}).\end{aligned} \tag{19}$$

This module is used for backpropagation in both sides A and B, filtering each loss function in (14)–(19).

IV. Experiments

In this section, we evaluate the effectiveness of our proposed method on two multiclass and one binary class classification datasets. The first two datasets, DeepGlobe landcover and Mangrove detection, are regarded as noisily annotated datasets due to specific reasons. The DeepGlobe dataset is a coarsely annotated public dataset used for a competition in 2018 [57], and it has been noted to contain noisy pixels due to the difficulty of landcover mapping [58]. The Mangrove dataset is used for monitoring mangrove areas which play crucial roles in coastal ecosystems. Previous research has pointed out the difficulty of classification due to surrounding similar terrestrial vegetation and changes in coastal areas due to periodic tides [14], [15], resulting in the risk of wrongly annotated labels. The last dataset is the Massachusetts dataset, which is a well-used public dataset for building extraction and is not regarded as noisy. However, we introduce artificial noise into the labels to create a simulated noisy setting following Liu et al. [40].

A. Datasets and Settings

1) DeepGlobe Landcover Dataset: The DeepGlobe landcover dataset was used for a public competition held in 2018 [57] and is regarded as one of the standard public datasets for Landcover classification. It has 803 high-resolution images, each with a size of 2448 × 2448 pixels, where seven classes of landscape regions—urban, agriculture, rangeland, forest, water, barren, and unknown—are defined. Following previous settings [58], [59], [60], we split a total of 804 images into 455 training, 207 validation, and 142 testing images, respectively, and the "unknown" class is not considered in the evaluation. Although this dataset includes a relatively large number of pixels of samples across multicities compared with other datasets, it integrates different visibility areas into only one simple class implicitly, resulting in label noise, as noted in [58]. In our experiments, we set the epochs to 80, the patch size to 512 × 512, the initial learning rate to 0.00008, and the batch size to 6 for PSPNet and DeepLabV3+, and 8 for other models with gradient accumulation to manage memory constraints. We use the Adam optimizer and an exponential learning rate scheduler with a gamma of 0.90, with an early stopping set at the highest metric score.

2) Mangrove Detection Dataset: The Mangrove detection dataset was created by the Queensland Herbarium and Biodiversity Science for specific purposes, such as the conservation of mangrove vegetation and biodiversity in Queensland, Australia. The categories follow community information from the source dataset [61], including mangrove communities,

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saltmarsh communities, *Casuarina glauca* communities, water, and others. The dataset consists of 652 high-resolution satellite images, which were split into 522 training and 130 testing images, each with a size of 256 × 256 pixels. Given that mangrove observation is often affected by various environmental factors, as discussed in Section II-C, the annotated labels commonly contain noisy information. Moreover, it is worth noting that this dataset is relatively small for a segmentation task, not due to coarse annotation processes, but rather due to its inherent difficulty, which distinguishes it from weakly annotated research fields such as WSSS. In our experiments, we set the epochs to 60, the patch size to 256 × 256, and the initial learning rate to 0.00001, with a batch size of 8. We use the SGD optimizer with a momentum of 0.999 and the same scheduler and early stopping settings as the DeepGlobe dataset.

3) Massachusetts Building Dataset: The Massachusetts dataset is a well-used public dataset and consists of 151 aerial images, each with dimensions of 1500 × 1500 pixels. These images were randomly divided into three subsets: a training set comprising 137 images, a test set containing ten images, and a validation set consisting of four images. For experiments on noisy labels, we inject incomplete label noise following the previous study [40], where the discarding percentage α_0 is set to 0.3. To replicate the settings described in [40], we also performed random cropping with patch sizes of 256 × 256 pixels, without removing nonlabeled samples and obtained 4110, 120, and 300 patch samples for the training, validation, and test datasets, respectively. In our experiments, we set the epochs to 40 with a batch size identical to those used for the Mangrove dataset. We use the same optimizer, scheduler, and early stopping settings as the DeepGlobe dataset.

4) Experimental Detail: In our study, we conducted two types of experiments to validate the effectiveness of our proposed method. The first one focused on improvements in benchmark accuracy using various backbone networks, and the second one involved comparisons with other existing methods that address noisy labels. As backbone networks, we selected the original UNet [51], UNet with a ResNet-18 backbone (ResUNet), FCN-8s [52] with a VGG16 backbone, PSPNet [53] with a ResNet-50 backbone, DeepLabV3+ [54] with a ResNet-50 backbone, Segformer-B2 [55], and ST-UNet [56] with the parameters pretrained on the Posdom dataset [62] with the same settings in the original paper. For the second experiment, we use co-teaching [10], ADELE [31], SFL [19], and AIO2 [40] for comparison. Co-teaching is a standard method of sample selection, while ADELE is a multiclass segmentation method. SFL and AIO2 are binary-class segmentation methods but optimized for RS tasks. Here, due to the sensitivity of RS data to scale factors, we omitted the multiscale consistency loss in the original ADELE. While these experiments are primarily conducted on the DeepGlobe and Mangrove datasets, we used the Massachusetts buildings dataset to assess the extent to which bias introduced by label noise distorts validation results on test datasets, which is discussed in Section V.

For our proposed method, we set α to 2, 2, and 3, β to 0.8, 0.95, and 0.8, for the DeepGlobe, Mangrove, and Massachusetts datasets, respectively. In addition, we set scale factor in MPM to 2 for the DeepGlobe dataset, and 1 for the other datasets. Regarding the parameters of CRF-Loss, we adopt the gated CRF-loss format in [43] and use the same parameters, except for the kernel weight w of 0.040.

For the evaluation, we chose the mean intersection over union (mIoU) as the base metric. We eliminated the class "unknown" from the DeepGlobe dataset and calculated a single IoU for the class "building" on the Massachusetts dataset, following the settings used in previous papers [40], [58], [59], [60].

B. Experimental Results

1) Network Dependency: The first experimental result, focusing on improvements to benchmark settings that are not enhanced by any noise-robust methods, is shown in Table I. According to the table, it is evident that our proposed method significantly improves benchmarks across all the datasets and various backbone networks. Our experiments reveal that the improvements on PSPNet and DeepLabV3+ are relatively large compared with those on FCN-8s and Segformer-B2, while the improvement observed with UNet is quite small. Given the background of the memorization effects [9], model degradation stems from overfitting to noisy labels. Therefore, when the network representation ability is limited and can only memorize a small proportion of noisy labels, the improvement from our noise filtering process is expected to be limited. Similarly, the improvement in ST-UNet, which uses a large backbone combining PreAct ResNet [62] and Vision Transformer blocks [63], is much higher than that in ResUNet. This finding suggests that larger networks are more sensitive to noise and necessitate serious consideration for handling label noise. However, network size alone does not fully explain the higher performance of PSPNet and DeepLabV3+ than FCN-8s and Segformer-B2, which suggests that factors other than network capacity and representation ability are at play.

2) Comparison to Other Methods: The comparison results with previous methods across all the datasets are shown in Table II. For a fair comparison, the evaluation of co-training-based approaches (co-teaching and our method) is conducted using a single network. From the table, it is clear that our methods achieve the highest scores across all the methods and datasets. While the most experimental results on the Massachusetts dataset show improvements in accuracy across various methods, except co-teaching, those on the multiclass datasets are severe, resulting in degradation compared with benchmark models, which implies the importance of the factor of class number in learning from noisy labels.

Co-teaching shows severe degradation compared with the benchmark method across all the datasets. This demonstrates that adapting instancewise sample selection methods to segmentation tasks is quite different from adapting them to image classification tasks. As mentioned in Section II-A, the sample selection method carries the risk of error accumulation, especially when the loss distribution of clean and noisy samples is quite similar. In segmentation tasks, the loss distribution for an instance is calculated by averaging pixelwise losses affected by both clean and noisy samples, making it difficult to distinguish clean instances from noisy ones. In addition, sample selection reduces the batch size and the number of samples, resulting in decreased training stability. Therefore, simply adapting

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TABLE I
EVALUATION RESULTS OF MIOUS (%) ON THREE DATASETS

Dataset	Backbone Network	Benchmark (A)	Ours (B)	B - A
DeepGlobe Landcover	**UNet [51]**	40.541%	40.579%	+0.038%
	FCN-8s [52]	66.972%	67.219%	+0.247%
	PSPNet [53]	62.126%	63.189%	+1.063%
	DeeplabV3+ [54]	67.161%	68.275%	+1.114%
	Segformer-B2 [55]	69.034%	69.576%	+0.542%
Mangrove detection	**UNet [51]**	56.163%	57.103%	+1.940%
	PSPNet [53]	60.178%	63.522%	+3.344%
	DeeplabV3+ [54]	60.709%	64.716%	+4.007%
Massachusetts building	**ResUNet [51]**	56.968%	58.605%	+1.637%
	ST-UNet [56]	52.203%	58.658%	+6.455%

TABLE II
COMPARISON RESULTS OF MIOU (%) TO PREVIOUS METHODS ON THREE DATASETS WITH THE BACKBONE OF DEEPLABV3+

Method	DeepGlobe Landcover	Mangrove detection	Massachusetts building	Averaged in 3 datasets
Benchmark	67.161%	60.709%	56.968%	61.613%
Co-teaching [10]	66.072%	59.459%	53.776%	59.769%
ADELE [31]	67.034%	63.110%	–	–
SFL [19]	64.879%	58.150%	57.370%	60.133%
AIO2 [40]	64.587%	58.896%	58.315%	60.599%
Ours	**68.275**%	**64.716**%	**59.769**%	**64.253**%

standard instancewise selection methods such as co-teaching to segmentation cases does not seem reasonable.

ADELE was designed for multiclass classification tasks, so we conducted experiments on only two multiclass datasets. This method shows effectiveness on the Mangrove dataset and does not exhibit apparent degradation on the DeepGlobe dataset. ADELE adopts an early learning strategy and replaces uncertain pixels with pseudolabels for the detected classes that seem to be overfitting, by fitting the curves of the metric function. Intuitively, this approach carries the risk of inaccurate fitting and may detect inappropriate points to start pixel correction, which may relate to learning settings such as the number of classes, as well as the content of datasets. We discuss this point in detail in Section V.

Both SFL and AIO2 demonstrate improvement over the benchmarks on the Massachusetts dataset as reported in the original papers; however, they exhibit severe degradation on the two multiclass datasets, even compared with co-teaching. SFL is a sample selection method that focuses on determining when to filter out samples and how many should be eliminated, whereas AIO2 is based on an early learning strategy similar to ADELE, with pixelwise corrections by model predictions. The lack of effectiveness in multiclass cases seems to be related to the specific features of binary classification; label noise is directly linked to the single target object itself, and models do not need to estimate complex flipping probabilities from one class to another. As a result, these strategies can be regarded as optimized methods for binary classification cases, but not for multiclass cases.

On the other hand, our method shows the best performance across all the datasets, which appears to address the aforementioned weakness. Our module, comprising MPM, supported by NLPM and LFM, can produce optimized masks for specific epochs and areas by learning from the model's errors on the side opposite to the input. This module seems to possess similar functions to the SFL modules or the early learning strategies used in AIO2, resulting in improved performance on the Massachusetts dataset. The effectiveness of our method on multiclass datasets stems from the loss filtering process in MPM; it extracts interclass and spatial relationships, as illustrated in Fig. 3, but does not rely on complex fitting processes or label correction by model predictions, which are inherently uncertain. The MPM simply suppresses the influence of uncertain pixels without introducing any additional information into the loss calculation, while the loss calculation in LFM includes compensation terms with CRF-Loss as in (18) for pixels with less information, assuming that estimating the appropriate class based on labels from surrounding clean pixels is an easier approach than relying on the uncertain pixels themselves. These strategies do not impose any strict limitations on incorporating existing networks and are expected to be practically useful for improving robustness against noisy labeled datasets, which seems to result in adaptability to multiclass classification, achieving the highest scores on the DeepGlobe and Mangrove datasets.

V. DISCUSSION

To further compare performance across methodologies, we use Cohen's kappa κ, ΔIoU, and the Top-3 mIoU for detail model evaluation. Cohen's kappa is used to assess the consistency between curators, while ΔIoU, and the Top-3 mIoU measure the gap between the maximum and minimum class IoU and assess the average in the top-3 class IoU, respectively, where the class selection is based on the benchmark results. The analysis results are summarized in Table III.

Cohen's kappa has been mathematically analyzed in relation to the label noise distribution in datasets annotated by multiple raters [64]. Because it is designed to subtract chance agreement among curators, it is expected to mitigate distortions

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TABLE III
ANALYSIS RESULTS ON THE DEEPGLOBE LANDCOVER AND MANGROVE DETECTION DATASETS WITH THE BACKBONE OF DEEPLABV3+

Method	DeepGlobe Landcover			Mangrove Detection		
	κ	ΔIoU	Top-3 mIoU	κ	ΔIoU	Top-3 mIoU
Benchmark	0.7721	52.863%	79.170%	0.7610	68.002%	73.140%
Co-teaching [10]	0.7656	55.752%	78.708%	0.7598	81.849%	75.064%
ADELE [31]	0.7699	54.856%	79.840%	0.7676	68.668%	76.866%
SFL [19]	0.7593	61.370%	79.495%	0.7467	78.319%	72.729%
AIO2 [40]	0.7566	58.156%	78.500%	0.7604	82.353%	74.607%
Ours	**0.7833**	**50.812%**	**79.918%**	**0.7862**	**67.527%**	**77.432%**

TABLE IV
VALIDATION OF ASSESSMENT WITH LMN ON MASSACHUSETTS BUILDING DATASET

Method	κ_{GT} (A)	κ_{LMN} (B)	B - A
Benchmark	0.6778	0.5615	-0.1163
Co-teaching [10]	0.6494	0.5489	-0.1005
SFL [19]	0.6811	0.5628	-0.1183
AIO2 [40]	0.6897	0.5691	-0.1206
Ours	0.7018	0.5752	-0.1266
mean	–	–	-0.1165
std	–	–	0.0091

in test evaluations due to noise, as discussed in detail in Section VI. From Table III, it is evident that our proposed method exhibits the highest kappa κ and demonstrates superior correspondence to test labels compared with other methods. Furthermore, upon comparing the mIoU evaluations in Table II, the rankings of methods in terms of mIoU align with the kappa-based results, enhancing the reliability of the evaluation in Table II.

To further investigate the validity of kappa evaluation, we created an additional test dataset with artificially introduced noise, termed labeled masks with noise (LMN), along with ground truth (GT) and calculate Cohen's kappa κ on the Massachusetts dataset, as shown in Table IV. Due to the significant proportion of random noise, κ_{LMN} showed a considerable score decrease compared with κ_{GT}, thus we subtracted κ_{GT} (A) from κ_{LMN} (B) and calculated $B - A$ to assess the noise influence for each method. Since the noise added to LMN does not vary by method, if the evaluation based on kappa is noise-robust, $B - A$ is expected to be independent of the method. To test this hypothesis, we calculated the statistical values of $B - A$ for each method as shown in Table IV and confirm that the standard deviation (0.0091) appears relatively small compared with the absolute value of the mean (0.1165). This result supports the robustness of the evaluation based on kappa to noise bias.

From ΔIoU and Top-3 mIoU, we can get valuable insights about model tendency. We can observe that our proposed method exhibits the lowest ΔIoU and highest Top-3 mIoU on the DeepGlobe and Mangrove datasets, indicating that the trained model does not exacerbate overfitting, as shown in Table III. Conversely, other methods show relatively high ΔIoU, while some of them improve Top-3 mIoU compared with benchmark results, suggesting that both the sample selection approaches (co-teaching and SFL) and early learning methods (ADELE and AIO2) tend to favor certain classes. For instance, ADELE shows significant improvement of the accuracy in the top-3 classes compared with the benchmark, but worsens the ΔIoU on the Mangrove datasets, as shown in Table III, meaning that it improves the accuracy of preferred classes at the expense of hard classes.

The background of weak class preference in our method stems from the two modules: MPM and NLPM. In MPM, the initial convolution and multiplication with the label map facilitate learning the relationships between classes, reflecting flipping probabilities from prediction classes to label classes, rather than solely depending on the difficulty of individual classes. In addition, NLPM introduces noise regions for teaching MPM, limited to statistically prominent pixels with high loss within the batch samples. Consequently, MPM is expected to train to produce probabilities of extremely high loss on each pixel, thereby suppressing the reflection of confidence levels for each class and allowing focus solely on abnormal pixels. Through the functionalities of these modules, our method achieves a lower ΔIoU even compared with the benchmark, indicating its capacity to reduce the risk of overfitting.

In Figs. 4–6, we present the predicted maps across all the methods for better interpretation. As depicted in Fig. 4, the benchmark model failed to capture the rangeland areas, indicated in the colored areas of magenta in both the upper and lower images; however, our proposed method managed to partially extract these areas correctly. Similarly, in Fig. 5, we observe that our method effectively suppressed misclassified areas labeled as "others" due to uncertainty, unlike the benchmark. While other previous methods showed improvements in certain regions, it appears to be at the expense of accuracy in other instances, as seen in (c) and (g) of both the figures. In the experiments on the Massachusetts datasets, our method partially extracted noise affected buildings, which are missed by the benchmark method due to a loss of responsiveness, as depicted in Fig. 6, demonstrating an improvement in robustness to incomplete types of noise in binary classification.

VI. ABLATION STUDY

To gain deeper insights into our method, we performed extensive analyses focusing on three key aspects. First, we visually examined the evolution of filtering masks during the early stages of training, comparing their metrics at each iteration with those of the model on the training dataset. Second, we conducted a sensitivity analysis on the hyperparameters α and β to highlight the challenges involved in tuning them. Finally, we investigated the noise robustness of

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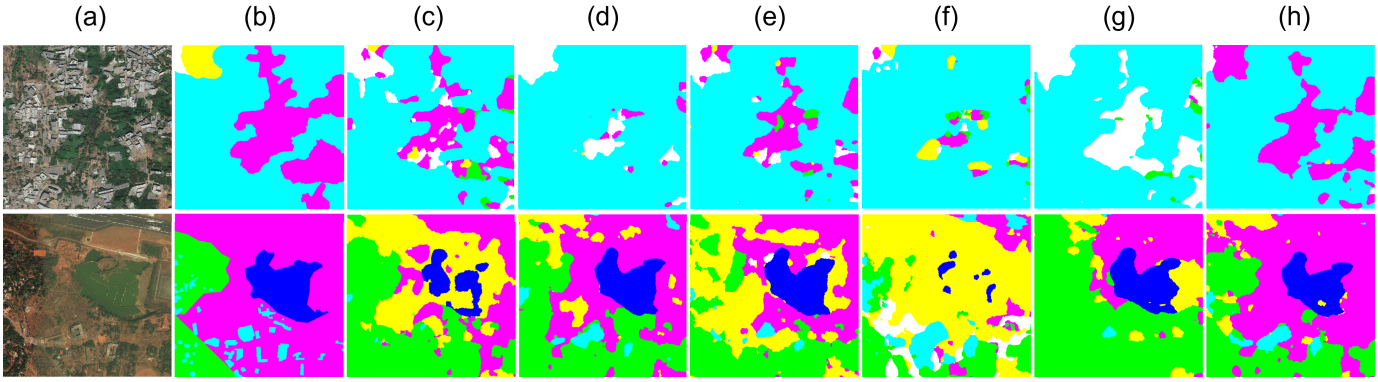


Fig. 4. Predicted results on the GeepGlobe dataset are shown for (a) input images, (b) label maps, (c) predicted maps from benchmark, (d) co-teaching, (e) ADELE, (f) SFL, (g) AIO2, and (h) our proposed method where the classes for "urban land," "agriculture," "rangeland," "forest land," "water," "barren land," and "unknown" are represented by cyan, yellow, magenta, green, blue, white, and black, respectively.

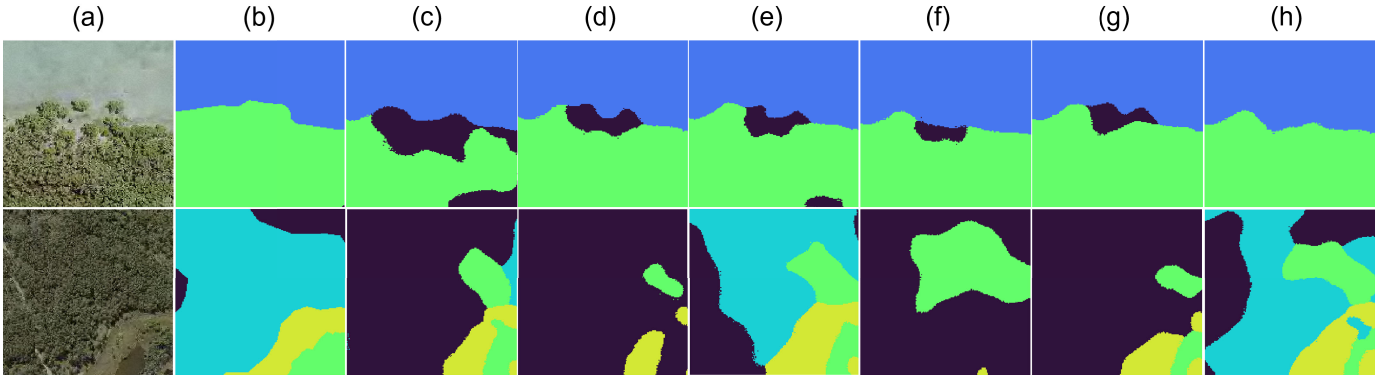


Fig. 5. Predicted results on the Mangrove dataset are shown for (a) input images, (b) label maps, (c) predicted maps from Benchmark, (d) co-teaching, (e) ADELE, (f) SFL, (g) AIO2, and (h) our proposed method where the classes for "mangrove communities," "saltmarsh communities," "*Casuarina glauca* communities," "water," and "others" are represented by green, yellow, cyan, blue, and black, respectively.

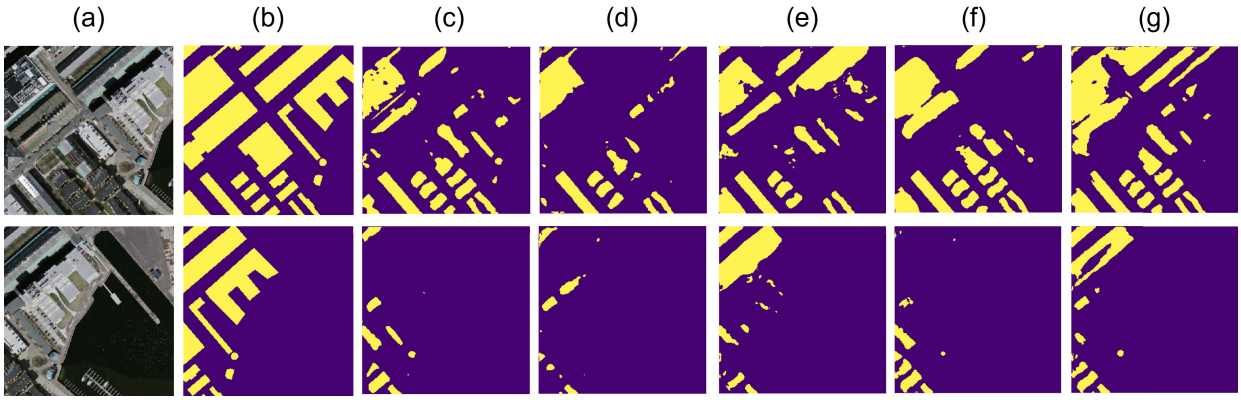


Fig. 6. Predicted results in the Massachusetts dataset, which include around 30% incomplete noise [40], are shown for (a) input images, (b) label maps, (c) predicted maps from benchmark, (d) co-teaching, (e) SFL, (f) AIO2, and (g) our proposed method.

the kappa coefficient as an evaluation metric through simple simulation analyses.

In our framework, the masks generated by the MPM are designed to effectively filter out noisy regions in the loss maps, serving as uncertainty-prediction maps. When our modules function as intended, the MPM gradually learns to identify error-prone regions as the two networks learn the training labels and reach a consensus. In Fig. 7, we illustrate the evolution of filtering masks around the deleted building footprints (representing label noise) over the first 1200 iterations on the Massachusetts dataset with DeepLabV3+. The graph shows that the values of the filtering masks on erroneous regions gradually increase relative to their surroundings, successfully differentiating noisy areas from clean backgrounds. Next, we tracked the progression of the error detection score of the filtering masks in the Mangrove dataset. For this analysis, we used the ROC-AUC as the evaluation metric, which does not rely on any threshold used for classifying regions as noisy or clean. To further relate these changes to the training process, we recorded IoU-based model prediction scores at

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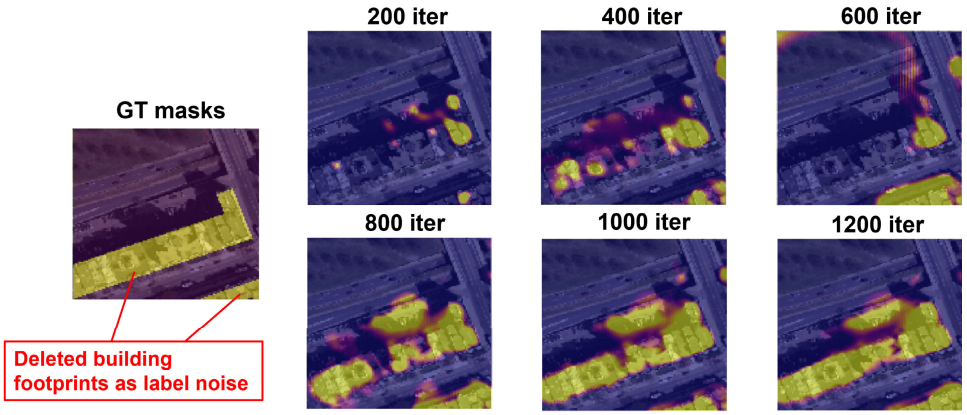


Fig. 7. Evolution of filtering masks over the initial 1200 iterations with a step size of 200 on the Massachusetts dataset. The left map displays the GT mask, representing a clean label map before any deletion noise is induced. The six heat maps on the right show the normalized filter masks produced by the MPM at each iteration.

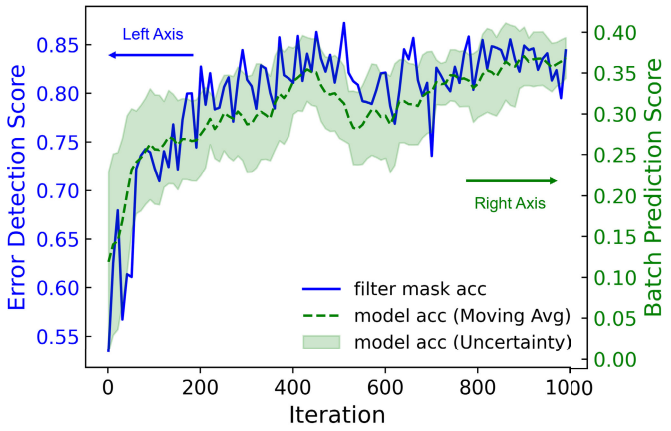


Fig. 8. Overlaid plot of mask error detection and model batch prediction scores during training on the Mangrove dataset. The left axis (error detection score) is based on the ROC-AUC, and the right axis (batch prediction score) is based on IoU. The model uncertainty, indicated by the green shaded area, is based on a range of ± one standard deviation.

TABLE V
SENSITIVITY ANALYSIS FOR α ON THE MANGROVE DATASET

α	0.0	1.0	2.0	3.0	4.0
mIoU (%)	62.59	64.63	64.72	64.60	63.53

each batch iteration and overlaid them with the ROC-AUC scores (see Fig. 8). To reduce fluctuations due to the limited sample size (8 samples per batch), we aggregated the results over ten iterations for the ROC-AUC evaluation and applied a moving average to the IoU scores, with uncertainty estimates based on their standard deviation. The overlapping trends in the two graphs indicate a strong correlation, suggesting that the MPM's error detection improves with model updates and, in turn, supports the training process during the early stages. This observation supports our explanation in Section III-C.

Regarding the sensitivity of hyperparameters in our framework, we varied the hyperparameters α and β over a wide range using a consistent step size on the Mangrove dataset for sensitivity analysis with DeepLabV3+. The evaluation results are summarized in Tables V and VI. For the hyperparameter α, which serves as the threshold for generating mask proposals

TABLE VI
SENSITIVITY ANALYSIS FOR β ON THE MANGROVE DATASET

β	0.0	0.2	0.4	0.6	0.8	1.0
mIoU (%)	62.52	63.01	63.72	64.01	64.19	62.73

in the NLPM from the loss maps, the performance remains stable within the range of 1.0–3.0 but degrades at 0.0 and 4.0. Interpreting α as the coefficient of σ in (13), this flat region is sufficiently wide for effective tuning. This stability can stem from the indirect nature of our loss filtering, where the MPM extracts consistent patterns from the proposal masks and optimizes the filtering regions during training. For β, which balances the contributions of the masked and unmasked terms in the CE-Loss within the LFM, we observe that increasing β generally leads to higher mIoU values—except at the maximum value of 1.0—indicating the effectiveness of our filtering process. The reduced performance at the upper bound likely reflects a tradeoff between noise reduction and information loss. In other words, if the filtering masks accurately detect uncertain regions, the process is beneficial; otherwise, it may result in the loss of valuable information. The upper bound might thus represent a limitation of the noise learning ability of our module. Overall, these results demonstrate that the two hyperparameters in the NLPM and LFM enable manual tuning of the balance and dependence of our filtering module without requiring overly sensitive adjustments.

To validate the use of Cohen's kappa coefficient κ, we next discuss its robustness to noise as an evaluation metric. Cohen's kappa κ subtracts the expected correct rate by chance—commonly represented as P_e—thereby eliminating contributions from random chance and promoting noise robustness. To validate this intuitive understanding, we conduct a simple simulation where we assume that $p\%$ of labels have been randomly flipped to one of the c classes. Assuming that the labels are well class-balanced, the observed accuracy P_o and Cohen's kappa κ on noisy labels can be related to the clean accuracy P_{cl} and clean kappa κ_{cl} as follows:

$$P_o \approx (1-p)P_{\text{cl}} + \frac{p}{c}$$
$$\kappa \approx (1-p)\kappa_{\text{cl}}.$$

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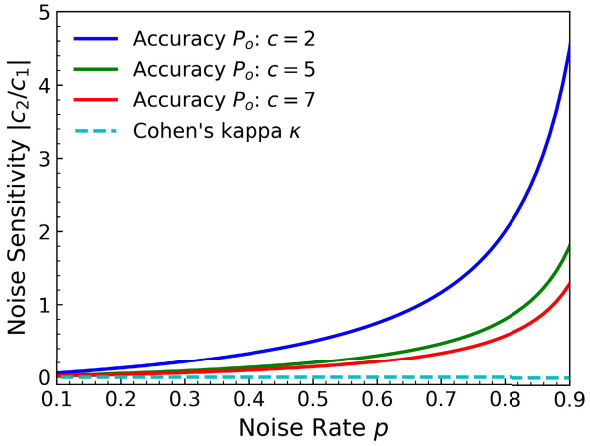


Fig. 9. Comparison of noise sensitivity $|c_2/c_1|$ between Cohen's kappa κ and accuracy P_o at each noise rate p across different numbers of classes c.

In this formulation, the only difference between the two metrics is the second term; therefore, we define the coefficients for the clean metrics as c_1 and c_2, respectively, and calculate the noise sensitivity as $|c_2/c_1|$. This ratio evaluates the extent of noise dominance in each metric on noisy labels. The simulated results in Fig. 9 show that when $p \approx 0$, both P_o and κ closely match their clean counterparts, resulting in $|c_2/c_1| \approx 0$. However, as the noise rate p increases, the accuracy P_o becomes increasingly affected by noise, resulting in the large value of $|c_2/c_1|$, whereas Cohen's kappa κ remains dependent solely on the clean terms, meaning $|c_2/c_1| \approx 0$. This simulation demonstrates that the formulation of Cohen's kappa effectively reduces the distortion caused by random noise and helps preserve the ranking of different methods. In real-world situations, noise can be more complex; however, given the random nature of noise, Cohen's kappa is one of the more robust evaluation metrics, thereby making the evaluation more reliable, supporting the discussion in Section V.

VII. Conclusion

In this article, we have introduced a method to handle noisy labels in remote sensing image segmentation. This method includes a co-training framework for reducing noise accumulation, a filtering module to mitigate the impact of loss function on uncertain pixels, and a weighted multiloss for compensating information by filtering operation. The experimental results demonstrate the effectiveness of our proposed method across various datasets and network architectures. We observed significant improvements in benchmark settings, particularly with PSPNet, DeepLabV3+, and ST-UNet, indicating the robustness of our approach. Comparison to previous methods revealed that our method outperforms others, with notable improvements in accuracy and robustness. The analysis of Cohen's kappa, Δ IoU, and Top-3 mIoU further supports the reliability and noise-robustness of our evaluation metrics. Our method's capacity to reduce overfitting and its ability to address weak class preferences highlight its effectiveness in real-world applications.

Despite the adaptability of our method, certain aspects remain unclear; the relationship between learning parameters (or network structure) and the type of noise is not fully understood. Furthermore, the potential of our method in scenarios involving asymmetric networks, where the two networks differ in their ability to extract features (e.g., receptive fields, feature resolutions, or levels of abstraction), remains unexplored. Addressing these gaps could be a promising direction for future work to deepen our understanding and further improve the efficacy of the proposed approach.

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